JEE Mathematics: Quick Review with Grading and Practice Map - Ellipse (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (60 questions; a question that appears in two of these is counted once): High = 5 or more questions, Medium = 1 to 4 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Ellipse: Introduction

Question 01

What is the condition for general equation of 2nd degree curve to represent an ellipse?

Model Answer

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is an ellipse if $\Delta \neq 0$ and $h^2 < ab$ (and $0 < e < 1$); circle if also $a = b,\ h = 0$.

Ex: $5x^2 - 2xy + 5y^2 - 76x - 88y + 506 = 0$: $h^2 = 1 < 25 = ab$, $\Delta \neq 0 \Rightarrow$ ellipse.

Grade - Low

Practice map, main concept: Sheet: J-I 7, 15; J-II 75, 95

Also used in: Sheet: C-I 20; C-II 65

Question 02

What is the definition of ellipse?

Model Answer

Locus of a point whose sum of distances from two fixed points (foci) is a constant $2a$, greater than the distance between them: $PS_1 + PS_2 = 2a > S_1S_2$.

Equivalently $PS = e \cdot PM$ with $0 < e < 1$ (focus S, directrix through M). If $2a = S_1S_2$ the locus is the segment S_1S_2.

Grade - Low

Practice map, main concept: DPP: 5(6, 12)

Also used in: none

Question 03

What is the equation of standard ellipse? and what is the relation between a and b?

Model Answer

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad b^2 = a^2(1 - e^2)$$

For the horizontal ellipse $a > b$ (major axis on x-axis); if $b > a$ the major axis is on the y-axis.

Grade - Medium

Practice map, main concept: Sheet: J-I 10; J-II 2; J-M 37, 54; J-A 2 | DPP: 8(7) | Question Bank: 110

Also used in: none

Question 04

Define major axis/minor axis, foci, vertices, center, principal axes for an ellipse.

Model Answer

Major (focal) axis: line through the foci, length $2a$, ends = vertices $(\pm a, 0)$. Minor axis: its perpendicular bisector, length $2b$, ends $(0, \pm b)$.

Foci $(\pm ae, 0)$; centre = meeting point of the axes (bisects every chord through it); the two axes together = principal axes.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 05

Define focal distance/focal radius, focal chord, double ordinate, latus rectum for an ellipse.

Model Answer

Focal distance: distance of a point of the ellipse from a focus. Focal chord: chord through a focus. Double ordinate: chord perpendicular to the major axis.

Latus rectum: the double ordinate through a focus (focal chord perpendicular to the major axis).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 42; C-II 30

Question 06

What is the formula of eccentricity?

Model Answer

$$e = \frac{ae}{a} = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \left(\frac{2b}{2a}\right)^2}, \qquad 0 < e < 1$$

e = (distance centre to focus) / (distance centre to vertex).

Ex: $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$: $e = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$.

Grade - High

Practice map, main concept: Sheet: J-II 81, 96; J-M 32, 56, 65 | DPP: 5(11); 8(4) | Question Bank: 98

Also used in: Sheet: J-II 10, 28, 82; J-M 6-7, 36-38, 44, 54, 67; J-A 2, 26, 30-31; C-II 3 | DPP: 5(7-8); 9(3, 18) | Question Bank: 97, 100, 102, 106

Question 07

Write expression for b in terms of a and e.

Model Answer

$$b^2 = a^2(1 - e^2) \quad \Leftrightarrow \quad a^2e^2 = a^2 - b^2$$

So the centre–focus distance ae satisfies $(ae)^2 = a^2 - b^2$; for a vertical ellipse take the bigger denominator as a^2.

Grade - Medium

Practice map, main concept: DPP: 5(17) | Question Bank: 109

Also used in: none

Question 08

What are the coordinates of ends of latus rectum and length of latus rectum for standard ellipse?

Model Answer

$$\left(\pm ae,\ \pm\frac{b^2}{a}\right), \qquad \ell_{LR} = \frac{2b^2}{a} = 2a(1-e^2)$$

Ex: $\frac{x^2}{25} + \frac{y^2}{9} = 1$: $e = \frac{4}{5}$, ends $\left(\pm 4, \pm\frac{9}{5}\right)$, $\ell_{LR} = \frac{18}{5}$.

Grade - High

Practice map, main concept: Sheet: J-II 93; J-M 44; J-A 26 | DPP: 5(8); 9(17) | Question Bank: 97, 102-103, 106

Also used in: Sheet: J-I 40; J-II 30, 57, 94; J-M 32, 57; C-I 18 | DPP: 5(5); 6(7); 9(16) | Question Bank: 95

Question 09

What is the relation between length of LR, e and distance between focus and corresponding directrix for an ellipse?

Model Answer

$$\ell_{LR} = 2e\,d, \qquad d = \frac{a}{e} - ae$$

d = distance of a focus from its directrix.

Semi-latus rectum = $e\times$(focus–directrix distance): use it when a focus and the directrix are given.

Grade - Medium

Practice map, main concept: Sheet: J-M 4

Also used in: none

Question 10

How to find focal distance of a point on an ellipse?

Model Answer

$$PS_1 = a - ex_1, \qquad PS_2 = a + ex_1, \qquad PS_1 + PS_2 = 2a$$

($S_1(ae, 0)$, $P(x_1, y_1)$ on the ellipse.) $PS_1 \cdot PS_2 = a^2 - e^2x_1^2$: least b^2 (at the vertices), greatest a^2 (at the ends of the minor axis).

Ex: $\frac{x^2}{25} + \frac{y^2}{16} = 1$, $P\left(3, \frac{16}{5}\right)$, $e = \frac{3}{5}$: $PS_1 = 5 - \frac{9}{5} = \frac{16}{5}$, $PS_2 = \frac{34}{5}$.

Grade - High

Practice map, main concept: Sheet: J-II 94; J-M 7 | Question Bank: 92, 99-100

Also used in: Sheet: J-M 58

Question 11

How to check whether two ellipses are similar?

Model Answer

Two ellipses are similar iff they have the same eccentricity (same ratio b/a).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 12
What is the effect on shape of ellipse if we increase eccentricity?

Model Answer
As e increases, $b/a = \sqrt{1-e^2}$ decreases: the ellipse becomes flatter (foci move towards the vertices). $e \to 0$: circle; $e \to 1$: segment.

Grade - Low
Practice map, main concept: none
Also used in: none

Auxiliary Circle and Eccentric Angle

Question 13
Define auxiliary circle for an ellipse.

Model Answer
The circle drawn on the major axis as diameter.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 14
What is the equation of auxiliary circle for standard ellipse?

Model Answer

$$x^2 + y^2 = a^2$$

Centre = centre of the ellipse, radius = semi-major axis (also for a vertical ellipse with a the bigger semi-axis).

Grade - Medium
Practice map, main concept: Sheet: J-I 39 | DPP: 7(11) | Question Bank: 94
Also used in: Sheet: J-A 38; C-II 1 | DPP: 5(13); 6(10)

Question 15
What are parametric coordinates of a point on standard ellipse?

Model Answer

$$P(\theta) = (a\cos\theta,\ b\sin\theta), \quad 0 \le \theta < 2\pi$$

Use it for max/min, loci and distances: one variable θ instead of (x, y). Ex: max of $x + y$ on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\sqrt{a^2 + b^2}$.

Grade - High
Practice map, main concept: Sheet: J-II 9-10; J-M 19, 36, 39; J-A 30, 36; C-II 65 | Question Bank: 104
Also used in: Sheet: J-II 96 | DPP: 10(2)

Question 16
What is eccentric angle of a point on ellipse?

Model Answer
For $P(a\cos\theta, b\sin\theta)$, θ is the angle made with the major axis by CQ, where $Q(a\cos\theta, a\sin\theta)$ is the corresponding point on the auxiliary circle (not the angle of CP).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 17
Define corresponding points.

Model Answer
P on the ellipse and Q on the auxiliary circle on the same ordinate (same side of the major axis): $P(a\cos\theta, b\sin\theta)$, $Q(a\cos\theta, a\sin\theta)$; $\dfrac{PN}{QN} = \dfrac{b}{a}$.

Grade - Medium
Practice map, main concept: Sheet: J-A 38
Also used in: Question Bank: 107

Question 18
How to find the locus when a point has coordinates of the form $(p\cos\theta,\ q\sin\theta)$ - e.g. a point dividing a rod sliding on the axes, or dividing the ordinate of a circle in a fixed ratio?

Model Answer
Write $h = p\cos\theta$, $k = q\sin\theta$ and eliminate θ: $\dfrac{h^2}{p^2} + \dfrac{k^2}{q^2} = 1$ (an ellipse).
Ex: rod of length $a + b$ with ends on the axes, P divides it in ratio $b : a$ from the x-axis: $P(a\cos\theta, b\sin\theta)$, locus $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$.

Grade - Medium
Practice map, main concept: Sheet: J-II 3, 28 | Question Bank: 107
Also used in: Sheet: J-II 96

Standard Ellipses

Question 19
What does 'everything' mean for an ellipse?

Model Answer
Centre, vertices, ends of minor axis, e, foci, feet of directrices, directrices, ends of latus rectum, length of latus rectum and auxiliary circle.

Grade - Medium
Practice map, main concept: DPP: 7(13) | Question Bank: 105
Also used in: Sheet: J-M 4 | DPP: 8(15)

Question 20
Find everything for standard ellipse.

Model Answer

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\ (a>b)$$

Centre $(0,0)$; vertices $(\pm a,0)$; minor ends $(0,\pm b)$; foci $(\pm ae,0)$; directrices $x=\pm\frac{a}{e}$; LR ends $\left(\pm ae,\pm\frac{b^2}{a}\right)$; $\ell_{LR}=\frac{2b^2}{a}$; aux. circle $x^2+y^2=a^2$.

Ex: $\frac{x^2}{169}+\frac{y^2}{144}=1$: $e=\frac{5}{13}$, foci $(\pm5,0)$, directrices $x=\pm\frac{169}{5}$, $\ell_{LR}=\frac{288}{13}$.

Grade - **High**

Practice map, main concept: Sheet: J-I 12, 14; J-II 57, 82, 91; J-M 38, 59; J-A 14, 19; C-II 29, 44 | DPP: 8(5, 15); 9(18)

Also used in: Sheet: J-I 9; J-M 65 | Question Bank: 94

Question 21
Find everything for standard ellipse (vertical).

Model Answer

$$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1\ (a>b)$$

Vertices $(0,\pm a)$; minor ends $(\pm b,0)$; foci $(0,\pm ae)$; directrices $y=\pm\frac{a}{e}$; LR ends $\left(\pm\frac{b^2}{a},\pm ae\right)$; $e^2=1-\frac{b^2}{a^2}$.

Ex: $\frac{x^2}{16}+\frac{y^2}{25}=1$: $e=\frac{3}{5}$, foci $(0,\pm3)$, directrices $y=\pm\frac{25}{3}$, $\ell_{LR}=\frac{32}{5}$.

Grade - **Medium**

Practice map, main concept: Sheet: J-M 6

Also used in: none

Question 22
Plot both types of standard ellipse and mark everything for them.

Model Answer

Draw the axes, mark $(\pm a,0),(0,\pm b)$ (or $(0,\pm a),(\pm b,0)$), the foci inside on the major axis, the directrices outside the vertices at distance a/e, and the LR through each focus.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 23
Find everything for ellipse if equation is given in PS = e PM form.

Model Answer

Read focus S, directrix and e (make the right side a true perpendicular distance). Major axis = perpendicular from S to the directrix; $\frac{a}{e}-ae$ = focus–directrix distance; centre on the axis at distance ae from S away from the directrix.

Ex: focus $(6,7)$, directrix $x+y+2=0$, $e=\frac{1}{\sqrt{3}}$: $6\left[(x-6)^2+(y-7)^2\right]=(x+y+2)^2\Rightarrow 5x^2-2xy+5y^2-76x-88y+506=0$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 8; C-II 55 | DPP: 5(1, 9, 13)

Also used in: DPP: 5(12) | Question Bank: 105

Question 24
How is the equation of a shifted ellipse represented?

Model Answer

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre (h,k); axes along $y=k$ and $x=h$.

Grade - Low
Practice map, main concept: DPP: 5(3)
Also used in: DPP: 6(9)

Question 25
How to observe whether equation of a 2nd degree curve represents shifted ellipse?

Model Answer
No xy term, x^2 and y^2 coefficients of the same sign and unequal. Complete the squares to $A(x-h)^2+B(y-k)^2=K$: ellipse if K has the sign of A,B; point if $K=0$; no locus otherwise.
Ex: $2x^2+3y^2-8x-18y+35=K \Rightarrow 2(x-2)^2+3(y-3)^2=K$: ellipse if $K>0$, point if $K=0$.

Grade - Low
Practice map, main concept: Sheet: C-I 20 | DPP: 5(2)
Also used in: Sheet: J-II 75, 95

Question 26
Find everything for shifted ellipse.

Model Answer
Put $X=x-h$, $Y=y-k$, read everything for $\frac{X^2}{a^2}+\frac{Y^2}{b^2}=1$, then shift back.
Ex: $9x^2+25y^2-18x+100y-116=0 \Leftrightarrow \frac{(x-1)^2}{25}+\frac{(y+2)^2}{9}=1$: centre $(1,-2)$, $e=\frac{4}{5}$, foci $(5,-2),(-3,-2)$, directrices $x=\frac{29}{4}$, $x=-\frac{21}{4}$, $\ell_{LR}=\frac{18}{5}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 57 | DPP: 5(5); 6(10); 8(1)
Also used in: Sheet: J-II 9

Question 27
How to find equation of shifted ellipse if some details about it are given?

Model Answer
Centre = mid-point of the foci (or of the vertices); direction of the major axis from the foci/vertices; then a from vertices or $2a$, ae from foci, $b^2=a^2-a^2e^2$.
Ex: foci $(-5,2),(3,2)$, major axis 10: centre $(-1,2)$, $a=5$, $ae=4$, $b^2=9$: $\frac{(x+1)^2}{25}+\frac{(y-2)^2}{9}=1$.

Grade - Medium
Practice map, main concept: Sheet: J-I 9; J-II 1; J-M 9 | DPP: 5(10); 6(9) | Question Bank: 95
Also used in: Sheet: C-II 44 | DPP: 7(13)

Question 28

Two fixed circles C1 and C2 are such that C2 is contained in C1. What is the locus of center of a variable circle which touches C1 internally and C2 externally?

Model Answer

$CC_1 = r_1 - r$, $CC_2 = r + r_2 \Rightarrow CC_1 + CC_2 = r_1 + r_2$: an ellipse with foci C_1, C_2 and $2a = r_1 + r_2$.

Ex: $x^2 + y^2 = 9$ and $(x-2)^2 + y^2 = 1$: foci $(0,0), (2,0)$, $a = 2$, $b^2 = 3$: $\frac{(x-1)^2}{4} + \frac{y^2}{3} = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-M 5; C-II 3, 36

Also used in: none

Question 29

Find everything for ellipse if equation is given in PS1 + PS2 = 2a form.

Model Answer

S_1, S_2 = foci; centre = their mid-point; $2ae = S_1S_2$; a from $2a$; $e = \frac{S_1S_2}{2a}$; $b^2 = a^2 - a^2e^2$. Valid only if $2a > S_1S_2$.

Ex: $\sqrt{(x-3)^2 + y^2} + \sqrt{(x+3)^2 + y^2} = 10$: $a = 5$, $ae = 3$, $b^2 = 16$: $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

Grade - Low

Practice map, main concept: Sheet: J-II 80 | DPP: 5(7)

Also used in: none

Alternate Definition of Ellipse

Question 30

Derive alternate definition of ellipse.

Model Answer

$$\frac{p_2^2}{a^2} + \frac{p_1^2}{b^2} = 1$$

In $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $|x| = p_2$ = distance from the minor axis, $|y| = p_1$ = distance from the major axis; this holds for any pair of perpendicular lines as axes.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 31

Find everything for ellipse if equation is given in alternate definition form.

Model Answer

Write each bracket as a perpendicular distance (divide by $\sqrt{l^2 + m^2}$). The line whose distance is over the bigger denominator (a^2) is the minor axis; centre = meeting point of the two lines.

Ex: $\frac{(x-y+1)^2}{32} + \frac{(x+y-3)^2}{50} = 1 \Rightarrow \frac{p^2}{16} + \frac{q^2}{25} = 1$: $a^2 = 25$, $b^2 = 16$, $e = \frac{3}{5}$, $\ell_{LR} = \frac{32}{5}$, centre $(1, 2)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Position of a Point w.r.t. Ellipse

Question 32

Which region is called inside/outside w.r.t. an ellipse?

Model Answer

Inside: the region containing the centre (and foci). Outside: the region containing points from which two real tangents can be drawn.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 33

Write S, S1 and T for standard ellipse.

Model Answer

$$S = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1, \quad S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1, \quad T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 34

How to find relative position of a point w.r.t. an ellipse?

Model Answer

$S_1 < 0$: inside; $S_1 = 0$: on; $S_1 > 0$: outside (write the equation with RHS 0 and leading terms positive).

Ex: $(2, 1)$ and $\frac{x^2}{9} + \frac{y^2}{4} = 1$: $S_1 = \frac{4}{9} + \frac{1}{4} - 1 = -\frac{11}{36} < 0$, inside.

Grade - Low

Practice map, main concept: Sheet: J-I 36 | DPP: 6(1, 14-15); 7(4)

Also used in: none

Chord Joining Two Points

Question 35

Write expression of equation of chord joining two points $P(\alpha)$ and $Q(\beta)$.

Model Answer

$$\frac{x}{a}\cos\frac{\alpha+\beta}{2} + \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha-\beta}{2}$$

Chord through $(d, 0)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{d-a}{d+a}$; focal chord $(d = ae)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{e-1}{e+1}$.

Grade - Low

Practice map, main concept: Sheet: J-I 41; J-II 12; C-II 35, 37

Also used in: Sheet: C-II 33

Line and Ellipse

Question 36

What is condition of tangency for standard ellipse? and how is it used?

Model Answer

$y = mx + c$ touches $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \iff c^2 = a^2m^2 + b^2$

So $y = mx \pm \sqrt{a^2m^2 + b^2}$ is a tangent for every m (two parallel tangents). Use it for tangents of given slope, common tangents and loci.

Ex: tangents to $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with equal intercepts ($m = -1$): $c^2 = 25$, $x + y = \pm 5$.

Grade - **High**

Practice map, main concept: Sheet: J-I 28, 34; J-II 31, 43; J-M 35, 66-67; J-A 41; C-II 28, 33 | DPP: 6(3, 5); 7(8) | Question Bank: 96

Also used in: Sheet: J-II 29, 32, 46; J-M 9, 17; J-A 15; C-II 32 | Question Bank: 108

Question 37

How to apply condition of tangency for different type of standard ellipses?

Model Answer

$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ (vertical): $c^2 = b^2m^2 + a^2$ - always $c^2 =$ (denominator of x^2)$\cdot m^2 +$ (denominator of y^2).

$lx + my + n = 0$ touches $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ iff $a^2l^2 + b^2m^2 = n^2$; $x\cos\alpha + y\sin\alpha = p$ iff $p^2 = a^2\cos^2\alpha + b^2\sin^2\alpha$. Shifted ellipse: shift the line too.

Grade - **Low**

Practice map, main concept: DPP: 6(2); 7(14)

Also used in: Sheet: J-I 34

Director Circle

Question 38

What is equation of director circle for a standard ellipse?

Model Answer

$$x^2 + y^2 = a^2 + b^2$$

Locus of points from which the two tangents are perpendicular; radius = distance from a vertex to an end of the minor axis.

Ex: tangents from any point of $x^2 + y^2 = 25$ to $9x^2 + 16y^2 = 144$ are at $90°$ (since $16 + 9 = 25$).

Grade - **Low**

Practice map, main concept: Sheet: J-I 37; J-II 15; C-II 24 | DPP: 7(2); 9(3)

Also used in: Sheet: J-II 97 | DPP: 6(10)

Question 39

Derive equation of director circle of standard ellipse.

Model Answer

$y = mx + \sqrt{a^2m^2 + b^2}$ through (h, k): $(h^2 - a^2)m^2 - 2hkm + k^2 - b^2 = 0$. Perpendicular: $m_1m_2 = \frac{k^2-b^2}{h^2-a^2} = -1 \Rightarrow h^2 + k^2 = a^2 + b^2$.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 40

How to find locus of a variable external point P for which condition on slope of tangents drawn from P to ellipse is given?

Model Answer

$$m_1 + m_2 = \frac{2hk}{h^2 - a^2}, \qquad m_1 m_2 = \frac{k^2 - b^2}{h^2 - a^2}$$

Put these in the given condition and replace (h, k) by (x, y). Angle θ between tangents: $\tan\theta = \frac{|m_1 - m_2|}{|1 + m_1 m_2|}$.

Ex: $\cot\theta_1 + \cot\theta_2 = k^2$ (angles with the major axis): $\frac{m_1+m_2}{m_1 m_2} = k^2 \Rightarrow 2xy = k^2(y^2 - b^2)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Tangent & Normal

Question 41

How to find equation of tangent for an ellipse if point of tangency is given?

Model Answer

$$T = 0: \quad \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

Ex: at $(3\sqrt{3}, 1)$ on $\frac{x^2}{36} + \frac{y^2}{4} = 1$: $\frac{\sqrt{3}x}{12} + \frac{y}{4} = 1$, i.e. $\sqrt{3}x + 3y = 12$.

Grade - High

Practice map, main concept: Sheet: J-A 7, 20, 23, 31; C-I 18; C-II 31 | DPP: 6(4, 7); 9(16)

Also used in: Sheet: J-II 81; J-A 38, 42

Question 42

How to find equation of tangent for an ellipse if any other condition other than point of tangency is given?

Model Answer

Take the slope form $y = mx \pm \sqrt{a^2m^2 + b^2}$ (slope / parallel / perpendicular / through a point) or the parametric form $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$ (intercepts, areas, loci) and apply the condition.

Ex: $3x^2 + 4y^2 = 12$, parallel to $y + 2x = 4$: $m = -2$, $c^2 = 16 + 3$: $y = -2x \pm \sqrt{19}$.

Grade - Low

Practice map, main concept: Sheet: J-II 32

Also used in: none

Question 43

What is the equation of tangent at a point, whose coordinates are given in parametric form, on standard ellipse?

Model Answer

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

Intercepts $\frac{a}{\cos\theta}, \frac{b}{\sin\theta}$: least area of the triangle with the axes $= ab$ (at $\theta = \frac{\pi}{4}$); least intercept between the axes $= a + b$.

Locus of the mid-point of the intercept: $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$.

Grade - Medium

Practice map, main concept: Sheet: J-II 46; J-M 8, 18, 20; C-I 11-12; C-II 1, 25, 27, 32 | DPP: 5(14-16); 6(6, 8); 7(3, 5)
Also used in: Sheet: J-I 37; J-II 43, 54, 98; C-II 4, 38, 64

Question 44
How to find equation of tangent to standard ellipse passing through an external point?

Model Answer
Put (h, k) in $y = mx + c$, $c^2 = a^2m^2 + b^2$: $(h^2 - a^2)m^2 - 2hkm + k^2 - b^2 = 0$; two roots m = two tangents (if $h^2 = a^2$ one tangent is vertical, $x = h$). Or use $SS_1 = T^2$.
Ex: $9x^2 + 16y^2 = 144$ from $(2, 3)$: $(3 - 2m)^2 = 16m^2 + 9 \Rightarrow m = 0, -1$: $y = 3$, $x + y = 5$.

Grade - Medium
Practice map, main concept: Sheet: J-II 59-60; J-M 55; J-A 42
Also used in: none

Question 45
How to find point of tangency if equation of tangent is given?

Model Answer

$$y = mx + c \ (c^2 = a^2m^2 + b^2): \quad \left(-\frac{a^2m}{c}, \ \frac{b^2}{c}\right)$$

Or compare $lx + my = n$ with $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$: $(x_1, y_1) = \left(\frac{a^2l}{n}, \frac{b^2m}{n}\right)$.
Ex: $x + y = 3$ touches $x^2 + 2y^2 = 6$ at $\left(\frac{6}{3}, \frac{3}{3}\right) = (2, 1)$.

Grade - High
Practice map, main concept: Sheet: J-M 17; J-A 15 | DPP: 7(9) | Question Bank: 108
Also used in: Sheet: J-I 27; J-M 66; J-A 41 | Question Bank: 96

Question 46
How to find equation of normal for a standard ellipse if foot of normal is given?

Model Answer

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

Slope of normal = $\frac{a^2y_1}{b^2x_1}$ (negative reciprocal of the tangent slope $-\frac{b^2x_1}{a^2y_1}$).

Grade - Low
Practice map, main concept: Sheet: J-II 30
Also used in: none

Question 47
Write expression for equation of normal to standard ellipse in terms of eccentric angle θ.

Model Answer

$$ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$$

It meets the axes at $G\left(ae^2\cos\theta, 0\right)$ and $g\left(0, -\frac{a^2e^2}{b}\sin\theta\right)$; locus of the mid-point of Gg is an ellipse with the same e.

Grade - Low
Practice map, main concept: Sheet: J-I 38, 40; C-II 2, 22-23, 64 | DPP: 10(2)
Also used in: DPP: 6(6)

Question 48
How to find the condition for a line to be a normal to standard ellipse?

Model Answer

$y = mx + c$ is a normal $\iff c^2 = \dfrac{m^2(a^2 - b^2)^2}{a^2 + b^2 m^2}$

Compare the line with $ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$ and eliminate θ. At most four normals can be drawn from a point to an ellipse.

Grade - Low

Practice map, main concept: DPP: 9(11)

Also used in: none

Question 49
How to find point of intersection of tangents drawn at $P(\alpha)$ and $Q(\beta)$ on standard ellipse?

Model Answer

$$\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} \right)$$

Solve the two tangents, or use: PQ is the chord of contact of the meeting point. If $\alpha+\beta$ is constant, the meeting point moves on a line through the centre.

Grade - Medium

Practice map, main concept: Sheet: J-II 97

Also used in: Sheet: J-A 7

Question 50
How to find minimum distance between a point and standard ellipse?

Model Answer

The shortest distance is along the normal through the point: write the normal at θ through the point, or minimise $(a\cos\theta - h)^2 + (b\sin\theta - k)^2$ (or $(x-h)^2 + y^2$ with y^2 from the ellipse, $|x| \le a$).

Ex: largest circle with centre $(1, 0)$ inside $x^2 + 4y^2 = 16$: $(x-1)^2 + \frac{16-x^2}{4}$ is least at $x = \frac{4}{3}$: $r^2 = \frac{11}{3}$.

Grade - Low

Practice map, main concept: Sheet: J-II 33

Also used in: none

Question 51
How to find minimum distance between a line and standard ellipse?

Model Answer

If the line misses the ellipse: draw the tangent parallel to it (on the near side); the answer is the distance between these parallel lines and the nearest point is its point of contact.

Ex: $x^2 + 2y^2 = 6$ and $x + y = 7$: tangent $x + y = 3$, distance $\frac{4}{\sqrt{2}} = 2\sqrt{2}$, nearest point $(2, 1)$.

Grade - Low

Practice map, main concept: Sheet: J-I 27

Also used in: none

Question 52
What is the position of a line w.r.t. ellipse corresponding to number of common points between them?

Model Answer

$$(a^2m^2+b^2)x^2+2a^2mcx+a^2(c^2-b^2)=0$$

$c^2<a^2m^2+b^2$: secant (2 points); $c^2=a^2m^2+b^2$: tangent; $c^2>a^2m^2+b^2$: no common point. For chords use the roots of this quadratic (sum/product).

Grade - Medium
Practice map, main concept: Sheet: J-M 34 | Question Bank: 93
Also used in: none

Common Concepts

Question 53
Write expression for equation of chord whose mid point is given for standard ellipse.

Model Answer

$$T=S_1: \quad \frac{xx_1}{a^2}+\frac{yy_1}{b^2}=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}$$

Ex: $2x^2+5y^2=20$ bisected at $(2,1)$: $4x+5y=13$. Loci of mid-points of chords through a point / of given slope: put the condition into $T=S_1$.

Grade - Medium
Practice map, main concept: Sheet: J-II 49 | DPP: 7(1, 6) | Question Bank: 101
Also used in: none

Question 54
Write expression for equation of chord of contact for standard ellipse.

Model Answer

$$T=0: \quad \frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$$

Ex: from $(3,2)$ to $x^2+4y^2=9$: $3x+8y=9$.

Grade - Medium
Practice map, main concept: Sheet: J-II 61; J-M 58
Also used in: Sheet: J-II 59; C-II 24

Question 55
How to find mid point of a chord of standard ellipse whose equation is given?

Model Answer
Compare $lx+my=n$ with $T=S_1$: $\frac{h/a^2}{l}=\frac{k/b^2}{m}=\frac{h^2/a^2+k^2/b^2}{n}$.
Ex: chord $3x+8y=9$ of $x^2+4y^2=9$: mid-point $\left(\frac{27}{25},\frac{18}{25}\right)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 56
How to find external point for a chord of contact for standard ellipse if equation of chord of contact is given?

Model Answer
Compare $lx + my = n$ with $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$: $(x_1, y_1) = \left(\frac{a^2 l}{n}, \frac{b^2 m}{n}\right)$ (the pole of the line).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 57
Write expression for combined equation of pair of tangents for standard ellipse.

Model Answer

$$SS_1 = T^2$$

Use for the angle between the tangents or their intercepts on a line; for separate equations the slope method is shorter.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 55

Question 58
Write expression for equation of polar for standard ellipse.

Model Answer

$$T = 0: \quad \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

Polar of (x_1, y_1) = locus of the meeting points of tangents at the ends of chords through (x_1, y_1).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 59
What does T = 0 represent for a point, according to its position relative to standard ellipse?

Model Answer
Point on the ellipse: tangent at it. Point outside: chord of contact (= its polar). Point inside: its polar (a line not meeting the ellipse, parallel to the chord bisected at the point).

Grade - Low
Practice map, main concept: none
Also used in: none

Question 60
How to find coordinates of pole of a given line?

Model Answer

$$\left(-\frac{a^2 l}{n}, -\frac{b^2 m}{n}\right)$$

This is the pole of $lx + my + n = 0$.
Compare the line with $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.

Grade - Low

Practice map, main concept: none
Also used in: none

Question 61
From any variable point on a fixed line, tangents are drawn to the ellipse; then the chord of contact will pass through which point?

Model Answer
Through the pole of the fixed line: for $lx + my + n = 0$, the point $\left(-\frac{a^2 l}{n}, -\frac{b^2 m}{n}\right)$.

Ex: points on the directrix $x = 4$ of $\frac{x^2}{4} + \frac{y^2}{3} = 1$: chord $x + \frac{ky}{3} = 1$ always passes through the focus $(1, 0)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 62
Derive equation of diameter for standard ellipse.

Model Answer

$$y = -\frac{b^2}{a^2 m}\, x$$

Mid-point (h, k) of a chord of slope m: from $T = S_1$, slope $= -\frac{b^2 h}{a^2 k} = m$; so the mid-points lie on this line through the centre. Conjugate diameters: $m_1 m_2 = -\frac{b^2}{a^2}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Highlights

Question 63
What is the sum of focal distances of any point on ellipse?

Model Answer

$$PS_1 + PS_2 = 2a$$

Use it in triangles PS_1S_2: perimeter $= 2a + 2ae$; with $\angle S_1 P S_2 = 90°$, area $= b^2$.

Grade - Low
Practice map, main concept: Sheet: J-I 42; J-II 54; C-II 30 | DPP: 5(4)
Also used in: Sheet: J-II 55

Question 64
What is the product of length of perpendiculars drawn from foci on any variable tangent to the ellipse?

Model Answer

$$p_1 p_2 = b^2$$

Ex: $\frac{x^2}{5} + \frac{y^2}{3} = 1$: $(S_1F_1)(S_2F_2) = 3$.

Grade - Low
Practice map, main concept: Sheet: J-II 29, 98; C-II 26, 63 | DPP: 7(7, 12)
Also used in: Sheet: C-II 64

Question 65
What angle does portion of tangent intercepted between directrix and point of tangency subtend at focus?

Model Answer

$$90^\circ$$

So the circle on this portion as diameter passes through the corresponding focus.

Grade - **Low**
Practice map, main concept: DPP: 7(15)
Also used in: none

Question 66
What is the relation between auxiliary circle and circle drawn on any focal distance as diameter?

Model Answer
They touch each other internally.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 67
Where does foot of perpendicular from focus on any tangent to ellipse lie?

Model Answer
On the auxiliary circle $x^2 + y^2 = a^2$.
(From the centre instead: locus $(x^2 + y^2)^2 = a^2x^2 + b^2y^2$.)

Grade - **Low**
Practice map, main concept: Sheet: C-II 38 | DPP: 7(10)
Also used in: Sheet: J-I 28 | DPP: 7(11-12)

Question 68
Explain reflection property of Ellipse.

Model Answer
The normal at P bisects $\angle S_1PS_2$ (tangent bisects the external angle): a ray from one focus is reflected through the other focus.
Ex: ray from $(-4, 0)$ to P on $9x^2 + 25y^2 = 225$ reflects through $(4, 0)$; image of one focus in a tangent lies on the line through P and the other focus.

Grade - **Low**
Practice map, main concept: Sheet: J-II 55; C-II 4, 34 | DPP: 7(16)
Also used in: DPP: 7(10)